Student Homework Sheet — Stage 11: Evaluation & Risk Communication

All paths below are inside `homework/homework11/`.

Chain

In the lecture, we learned how to compare **gaussian-based vs bootstrap-based CIs**, run **scenario sensitivity**, and present **assumption-aware** results with subgroup checks.
Now, you will adapt those methods to your dataset to quantify uncertainty, compare at least two assumptions, and communicate risks clearly.

Task Overview

You will:

1. Run a **bootstrap** to estimate uncertainty for a prediction or metric.
2. Compare **two "what-if" scenarios** (e.g., imputation choice, distributional assumption, or model configuration).
3. Produce **visuals** (CI bands, error bars, residual plots, side-by-side scenarios, etc.).
4. Write a **stakeholder-facing summary** stating assumptions, risks, and how results change.

Step-by-step

1. Copy `stage11_evaluation-risk-communication_homework-starter.ipynb` into `homework/homework11/`, rename it `homework11_evaluation-risk-communication_submission.ipynb`, and open it.

2. **Load data & model**
   - Use your dataset or the provided synthetic fallback.
   - Starter files:

     - `homework11_evaluation-risk-communication_submission.ipynb`
     - `src/evaluation.py` (helper functions)
     - `data/raw/data_stage11_eval_risk.csv` (auto-created if missing)

3. **Baseline fit**
   - Fit your model and compute at least one metric (RMSE, MAE, AUC, etc.).

4. **Bootstrap**
   - Resample your data ≥ 500 times to estimate a **confidence interval** for the chosen metric or prediction.
   - Visualize uncertainty with **CI bands, error bars, or other relevant plots**.

5. **Scenario comparison (≥ 2)**
   - Examples: mean vs median imputation, Gaussian vs fat-tailed errors, drop vs fill, linear vs polynomial fit.
   - Show how results shift and discuss the implications.

6. **Subgroup diagnostics**
   - Pick a categorical split (e.g., `segment`) and compare residuals/metrics.
   - Identify hidden failures or confirm stability across subgroups.

7. **Visualize**
   - Side-by-side panels for scenarios and subgroups with captions: *Assumptions* and *Takeaway*.
   - Ensure axes are consistent for comparability.

8. **Write-up (≤ 1 page)**
   - State assumptions, describe risks, summarize sensitivity results.
   - Clearly communicate **when the model is reliable vs risky**.
   - Use plain language for stakeholders, e.g., *"Prediction holds if weekly volatility stays within X; model sensitive to missing-rate > 10%; Segment C underperforms."*

Deliverables

- `homework11_evaluation-risk-communication_submission.ipynb` containing:
- Bootstrap code & CI figure
- Scenario comparison visuals
- Subgroup diagnostics
- Markdown summary (assumptions, risks, sensitivity)
- Save any helper functions in `src/evaluation.py` if you modify or add functions.

Submission: Save your completed notebook in `homework/homework11/` and push to your repository.

Rubric (10 points)

- **Reproducibility (2 pts):** seed set, clear steps, functions reused.
- **Bootstrap CI (2 pts):** correct resampling & CI computation; interpretation is sound.
- **Scenarios (3 pts):** ≥ 2 clearly different assumptions; side-by-side visuals; consistent axes; labels.
- **Subgroup diagnostic (1 pt):** evidence of heterogeneity or confirmation of stability.
- **Stakeholder write-up (2 pts):** plain-language assumptions, "holds if…", "sensitive to…", risks and next steps.

Example Expectation

- A 2×2 grid: Baseline vs Scenario A vs Scenario B with CIs, plus a residuals-by-segment chart.
- A concise paragraph summarizing key insights for stakeholders.

Notes:
- Connect your homework to the lecture content: gaussian-based vs bootstrap-based CIs, scenario comparisons, and subgroup evaluation.
- Use all visuals and tables to communicate uncertainty and risk clearly.
- Document all assumptions and any deviations from starter notebook code.